

CS402: Advanced Artificial Intelligence & RAG Systems

Department of Computer Science | Fall Semester

Course Overview

This course explores foundational and modern principles of Artificial Intelligence, natural language processing, and neural retrieval systems. Students will delve into search algorithms, embeddings, dense vector retrieval, and Retrieval-Augmented Generation (RAG). Practical laboratory sessions will focus on FastAPI backend integration and LLM application development.

Instructor & Teaching Assistants

Lead Instructor: Dr. Sarah Jenkins (Email: s.jenkins@university.edu)
Office Hours: Tuesday & Thursday, 2:00 PM – 4:00 PM (Room 412, Turing Hall)
Teaching Assistant: Alex Rivera (Email: a.rivera@university.edu)
TA Lab Hours: Monday & Wednesday, 10:00 AM – 12:00 PM (CS Lab 3)

Grading Breakdown & Criteria

The course grading policy is structured to emphasize both theoretical understanding and hands-on implementation.

Assessment Category	Weight	Description
Programming Assignments (4x)	25%	FastAPI services, vector indexing, and chunking
Midterm Exam	20%	Foundational theory, search algorithms, probability
Course RAG Capstone Project	30%	End-to-end PDF Q&A RAG application with evaluation
Class Participation & Quizzes	10%	Weekly comprehension quizzes and active discussion
Final Examination	15%	Comprehensive theoretical and applied concepts

Late Submission & Attendance Policy

Assignments submitted after the deadline incur a 10% penalty per day up to a maximum of 3 days. Submissions beyond 72 hours past the deadline receive a grade of zero unless an official medical extension is approved prior to the due date. Each student is granted 2 slip days to be used throughout the semester for emergencies.

Weekly Schedule & Course Modules

Week	Topic	Core Readings & Deliverables
Week 1-2	Introduction to AI & Python Environment Setup	Search algorithms, heuristics (A*). Assignment 1 released.
Week 3-4	Vector Spaces & Text Embeddings	Cosine similarity, TF-IDF vs Dense vectors, semantic search.
Week 5-6	Vector Databases & Indexing Strategies	Chunking strategies, metadata tagging, ChromaDB/FAISS. Assignment 2.
Week 7	Midterm Review & Examination	Covers Weeks 1-6.
Week 8-9	Large Language Models & Prompt Engineering	Gemini API, zero-shot/few-shot, hallucinations and grounding.
Week 10-11	RAG Pipeline Architecture & Evaluation	Retrieval augmented generation, reranking, RAG triad. Assignment 3.
Week 12-13	FastAPI Production Deployment	Asynchronous endpoints, Dockerization, Swagger API docs.
Week 14	Capstone Project Presentations	Live demo of PDF RAG system and oral defense.

Capstone Project Requirements

For the 30% capstone project, students must build a complete RAG system that accepts PDF uploads, chunks and indexes the documents, and allows interactive Q&A; powered by an LLM. The project must be served via FastAPI with clean endpoint documentation at /docs. Citing exact page numbers and document sources is mandatory to prevent hallucinations.

Academic Integrity Policy

All code and deliverables must be the student's individual work. While using generative AI tools as a learning assistant is encouraged, copying another student's submission or directly copying unverified code constitutes academic misconduct and will be referred to the University Academic Board.